

Experiment 1

Code:

```
# FIND-S Algorithm without CSV (Google Colab)

# Training data
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

# Initialize hypothesis
hypothesis = ['0'] * (len(data[0]) - 1)

print("Training Data:")
for row in data:
    print(row)

# Apply FIND-S Algorithm
for row in data:
    if row[-1] == 'Yes':
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = row[i]
            elif hypothesis[i] != row[i]:
                hypothesis[i] = '?'

print("\nMost Specific Hypothesis:")
print(hypothesis)
```

Output:

... Training Data:

```
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes']  
['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes']  
['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No']  
['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
```

Most Specific Hypothesis:

```
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```
